

LAL BAHADUR SHASTRI INTL, India

WMO: 424790

Lat: 25.450N

Lon: 82.867E

Elev: 266

StdP: 14.56

Time Zone: 5.50 (IND)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	44.9	47.9	37.7	33.6	79.8	40.9	38.0	71.6	12.1	68.4	11.0	68.7	1.5	270	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	23.9	109.4	73.7	107.4	74.2	104.3	74.7	84.3	93.2	83.6	92.1	82.9	90.9	9.3	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.5	171.0	88.0	81.4	164.9	87.3	80.9	162.0	86.9	48.8	93.3	48.0	92.2	47.3	91.4	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.3	12.8	11.1	DB	40.8	112.9	3.0	1.4	38.6	113.9	36.9	114.8	35.2	115.6	33.1	116.6	
			WB	40.2	86.2	2.5	2.0	38.4	87.6	36.9	88.8	35.5	89.9	33.7	91.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.3	59.8	67.3	77.5	88.1	92.3	92.0	86.6	86.0	85.2	81.2	72.4	62.9
	DBStd	11.53	4.72	4.51	5.31	4.32	3.75	5.05	3.44	2.43	2.74	3.73	3.87	4.64
	HDD50	3	2	0	0	0	0	0	0	0	0	0	0	1
	HDD65	285	168	24	0	0	0	0	0	0	0	0	1	92
	CDD50	10707	306	485	852	1143	1313	1260	1135	1117	1055	968	673	400
	CDD65	5515	8	89	387	693	848	810	670	652	605	503	223	27
	CDH74	73379	86	843	4416	10152	13370	12554	8849	8417	7487	5320	1667	218
	CDH80	41318	7	204	2338	6581	9029	8296	4491	4042	3438	2382	497	13
Wind	WSAvg	3.8	2.9	3.4	4.0	4.8	5.7	5.6	4.7	4.3	4.1	2.4	1.9	2.3
Precipitation	PrecAvg	44.6	0.7	0.8	0.5	0.6	1.8	6.3	12.2	11.8	8.0	1.6	0.2	0.2
	PrecMax	57.0	2.2	2.9	2.3	2.0	4.0	15.4	19.6	17.2	14.4	6.3	2.0	2.2
	PrecMin	32.4	0.0	0.0	0.0	0.0	0.1	2.0	5.2	7.2	2.2	0.0	0.0	0.0
	PrecStd	6.5	0.6	0.8	0.5	0.4	1.0	3.4	3.4	2.8	3.2	1.5	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.4	89.8	102.6	110.9	111.5	111.6	102.1	97.1	97.0	95.3	89.8	82.1
		MCWB	63.2	67.4	69.7	68.8	73.3	75.7	81.5	82.8	82.1	78.3	71.9	64.3
	2%	DB	76.6	85.6	98.5	107.8	109.4	109.4	98.4	95.2	95.2	93.5	87.7	79.0
		MCWB	62.0	67.4	69.6	69.7	73.9	76.3	81.9	82.7	81.7	77.7	70.0	63.2
	5%	DB	73.4	82.4	95.1	105.6	107.4	106.2	95.3	93.4	93.5	91.8	85.7	76.8
		MCWB	60.1	65.4	68.9	70.2	74.5	78.2	81.9	82.6	81.5	76.8	69.3	62.3
	10%	DB	70.9	79.2	91.6	102.5	104.3	103.5	93.3	91.7	91.7	90.0	82.8	73.7
		MCWB	59.5	63.7	68.1	70.3	75.5	78.8	82.1	82.1	81.2	75.7	68.0	61.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.6	71.1	74.6	79.2	83.1	84.5	85.0	85.2	84.5	82.6	75.6	68.2
		MCDB	75.3	82.3	92.6	97.4	101.5	98.4	93.1	92.7	93.4	90.1	84.2	77.1
	2%	WB	64.5	69.1	72.1	76.9	81.4	83.4	84.1	84.2	83.5	81.0	73.3	66.0
		MCDB	71.8	81.3	90.8	96.8	98.9	96.6	92.2	91.9	91.5	88.8	82.9	75.1
	5%	WB	62.6	67.4	70.6	75.3	80.2	82.5	83.4	83.5	82.7	79.8	71.7	64.2
		MCDB	68.9	79.3	88.9	95.0	96.9	94.9	91.0	90.6	89.8	87.5	80.8	72.5
	10%	WB	61.0	65.7	69.3	73.6	79.0	81.6	82.7	82.7	81.9	78.8	70.0	62.6
		MCDB	68.5	76.0	87.1	92.6	95.4	93.7	89.8	89.3	88.6	86.1	79.7	70.5
Mean Daily Temperature Range	5% DB	MDBR	20.3	23.2	26.9	28.5	23.9	18.4	12.1	11.7	12.8	18.6	22.5	21.2
		MCDBR	24.8	26.3	30.2	31.4	27.5	23.7	15.5	13.9	15.6	19.6	23.3	24.6
		MCWBR	11.9	10.9	9.1	8.5	8.0	6.5	4.8	4.9	5.3	6.4	8.3	10.6
	5% WB	MCDBR	19.9	23.5	27.0	25.8	22.3	17.4	13.0	12.6	13.5	15.7	19.7	20.7
		MCWBR	11.2	10.9	9.4	9.0	8.0	5.9	5.2	5.3	5.3	5.9	8.3	10.5
Clear-Sky Solar Irradiance	taub		0.571	0.530	0.526	0.598	0.694	0.730	0.646	0.605	0.590	0.629	0.677	0.650
	taud		1.704	1.850	1.888	1.767	1.615	1.585	1.767	1.889	1.898	1.742	1.576	1.592
	Ebn at Noon		210	237	247	233	211	202	220	229	228	208	181	181
	Edh at Noon		67	62	62	72	84	86	72	63	61	68	76	73
All-Sky Solar Radiation	RadAvg		990	1416	1842	2068	2048	1742	1414	1472	1472	1444	1157	947
	RadStd		135	106	76	87	89	154	130	76	119	101	64	97

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page